Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/26/2011
PI Name:	Huff, Janice Ph.D.		
Project Title:	Impact of Radiation Quality on Cancer Processes in 2D and 3D Esophageal Cell Models		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	janice.l.huff@nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	281-483-9883
Organization Name:	Wyle Laboratories, Inc./NASA Johnson Space Center		
PI Address 1:	Wyle		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	11/15/2009	**End Date:**	09/01/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 9/1/2013 (from 6/30/2012) per PI (Ed., 10/18/2012) NOTE: End date is now 6/30/2012 (from 12/31/2010) per PI (2/11/2011) NOTE: End date changed to 12/31/2010 (from 11/14/2010), per S. Krenek/JSC (8/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rustgi, Anil (University of Pennsylvania)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Esophageal cancer is the 6th leading cause of cancer death worldwide and the 7th leading cause of cancer death in the U.S. Esophageal squamous cell carcinoma (ESCC) is the most common histological subtype of esophageal cancer, and is a rapidly metastasizing malignancy that is associated with poor patient prognosis (with a 5 year survival rate <20%). Esophageal cancer is associated with a wide variety of risk factors including ionizing radiation, as noted by its increased incidence as a secondary cancer after radiotherapy for primary carcinomas of the head and neck and mediastinal regions, as well as the high excess relative risk for ESCC observed in atomic bomb survivors. Because of the strong link between radiation exposure and the development of esophageal cancer and the high mortality associated with this type of malignancy, it is essential for cancer risk assessment in astronauts to understand the molecular mechanisms of radiation-enhanced carcinogenesis in the esophagus and to understand how radiation quality affects these mechanisms. The objective of this project is to investigate the effects of radiation on the development of esophageal cancer and specifically its influence on intercellular communication in the tissue microenvironment. Cues from the microenvironment can play a pivotal role in determining the fate of precancerous and transformed cells, mediating both tumor-suppressive and tumor-promoting effects through modulation of cell proliferation, cell death, migration, invasion, and recruitment of new blood vessels. We hypothesize that one path whereby radiation may impact esophageal carcinogenesis is through non-targeted mechanisms involving alterations in the stromal fibroblasts that drive preneoplastic changes in esophageal epithelial cells. Our model system consists of normal esophageal fibroblasts and immortalized esophageal epithelial cells and genetic variants grown using standard 2-D culture techniques. We also use a 3-D organotypic model that was developed in the laboratory of project Co-I Dr. Anil Rustgi at the University of Pennsylvania. In this tissue engineering model, esophageal epithelial cells are grown in the presence of fibroblasts in a realistic tissue-like environment that mimics the morphological features, differentiation markers, and growth characteristics observed in normal human tissue. In preliminary experiments, we observed an increase in epithelial cell migration and invasion, both of which are hallmarks of cancer, when esophageal epithelial cells were cultured in the presence of conditioned media from irradiated stromal fibroblasts. To characterize this observation and to establish methodologies to utilize 3-D organotypic culture techniques for radiobiology experimentation, we pursued the following subset of specific aims during the first year of this project: Aim 1 - Determine how radiation alters stromal fibroblast interaction with esophageal epithelial cells by measuring senescence, genome instability, and secretion of bioactive molecules. Aim 2 - Examine how radiation impacts hallmarks of carcinogenesis, specifically genomic instability, in 3-D organotypic cultures.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Understanding the early molecular mechanisms of carcinogenesis in the esophagus is important not only for risk assessment for astronauts who will encounter excessive amounts of radiation during long duration space travel, but also for applications here on Earth; these findings may contribute to advances in early disease detection and targeted treatment strategies for this rapidly fatal malignancy. In addition, contributions in the area of tissue engineering that result from the use of organotypic cell culture models may translate to and promote progress in other areas of biomedical research.		

Task Progress:	Aim 1 - Antibody-capture cytokine arrays (Raybiotech, Inc.) were used to profile bioactive compounds in the conditioned media of irradiated stromal fibroblasts. This study led to the identification of hepatocyte growth factor (HGF) as an abundant protein produced by esophageal fibroblasts that exhibited increased levels in media following irradiation with 2 Gy gamma-rays. HGF is a paracrine signaling molecule that works in concert with its cognate transmembrane tyrosine kinase receptor MET to mediate cell- and tissue-specific effects on a wide variety of cellular functions including cell proliferation, motility, invasion, cell survival and apoptosis. Using ELISAs, we quantified a dose-dependent increase in the levels of HGF in conditioned media of irradiated fibroblasts, and real-time qPCR was used to confirm increased mRNA levels for HGF at 1 hour post-radiation exposure. Purified, recombinant HGF alone caused an increase in the migratory capacity of epithelial cells in modified Boyden chamber assays and in scratch wound healing assays. Neutralizing antibodies against HGF blocked the enhanced migration and invasion of esophageal epithelial cells observed when esophageal epithelial cells were co-cultured with irradiated fibroblasts. Based on the results of this work we concluded that HGF was a major mediator of the increased migratory and invasive behaviors of esophageal epithelial cells observed in our preliminary work. We also conducted an initial characterization of the effects of HGF on genomic instability in esophageal epithelial cells using the cytokinesis-block micronucleus assay. Results of these studies showed an enhanced accumulation of genetically unstable cells in irradiated epithelial cells when cultured in the presence of HGF, suggesting a potential role for HGF in DNA repair or cell survival in this model. In summary, our Year 1 results suggest that intercellular communication through the HGF/MET axis plays an important role at multiple levels in the response of esophageal tissue to ionizing radiation. Aim 2 - To examine how radiation impacts carcinogenesis in 3-D organotypic cultures, we developed protocols for use of the cytokinesis-block micronucleus assay in epithelial cells harvested from organotypic cultures. These protocols were used to conduct a comparison study of the genotoxic effects of radiation on epithelial cells grown in standard 2-D monolayer culture compared to the same cells grown in 3-D organotypic culture. Data were collected using gamma-rays and high-LET silicon and titanium ion beams. For all radiation types, we observed an approximate 2-fold increase in genetic damage in cells grown under organotypic culture conditions compared to standard monolayer culture. One possible explanation for these results is that the increased cell contact afforded by 3-D culture offers protection from the cytotoxic effects of radiation in a manner similar to the phenomenon described as “multicellular resistance” where cancer cells grown as 3-D spheroids or in vivo exhibit an increased resistance to killing by chemotherapeutic agents compared to the same cells grown in 2-D. A variety of factors may be involved including increased cell-cell communication, microenvironment influences, and changes in cell cycle kinetics that may promote survival of damaged cells in 3-D that would otherwise die or be rendered reproductively inactive in 2-D culture. Further characterization of these data will be pursued during Year 2.
Bibliography Type:	Description: (Last Updated: 09/11/2019)
Abstracts for Journals and Proceedings	Patel ZS, Grugan KD, Rustgi AK, Cucinotta FA, Huff JL. "Modulation of Esophageal Epithelial Cell Behavior by Irradiated Stromal Fibroblast: Effects of Radiation Quality." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. Program and abstracts. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. p. 44. , May-2010
Abstracts for Journals and Proceedings	Patel ZS, Grugan KD, Rustgi AK, Cucinotta FA, Huff JL. "Influence of Ionizing Radiation on Stromal-Epithelial Communication in Esophageal Carcinogenesis." Committee on Space Research (COSPAR) 2010 38th Scientific Assembly, Bremen, Germany, July 18-25, 2010. COSPAR Abstract Book. Committee on Space Research (COSPAR) 2010 38th Scientific Assembly, Bremen, Germany, July 18-25, 2010. https://www.cospar-assembly.org/abstracted/COSPAR-10/abstracts/data/pdf/abstracts/E22-0023-10.pdf , Jul-2010
Abstracts for Journals and Proceedings	Patel ZS, Grugan KD, Rustgi AK, Cucinotta FA, Huff JL. "Effects of Ionizing Radiation on Stromal-Epithelial Communication in Esophageal Carcinogenesis." Presented at the 56th Annual Meeting of Radiation Research Society, Maui, HI, September 25-29, 2010. Abstract book. 56th Annual Meeting, Radiation Research Society, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=e50705ed-1fdf-4831-b075-cdfd85dfa8b3&cKey=48fffb64-0491-469a-b20f-1f412befc2a3&mKey={5FB93787-2553-4B3B-A915-8C1AE71F127A} , Sep-2010
Articles in Peer-reviewed Journals	Ponomarev AL, Huff J, Cucinotta FA. "The analysis of the densely populated patterns of radiation-induced foci by a stochastic, Monte Carlo model of DNA double-strand breaks induction by heavy ions." Int J Radiat Biol. 2010 Jun;86(6):507-15. PubMed PMID: 20470200 , Jun-2010